SEC 2-FA2 GROUP 3-CELLONA, BD; RICAFORT, NF-FA2

Github link: https://github.com/bryandioncellona/APM1110/blob/59892f23bd492ca988368d9154532a2933ebb98f/FA2/SEC%202-FA2%20GROUP%203-CELLONA%2C%20BD%3B%20RICAFORT%2C%20NF-FA2.Rmd

September 1, 2026

2. An experiment consists of tossing two fair coins. Use R to simulate this experiment 100 times and obtain the relative frequency of each possible outcome. Hence, estimate the probability of getting one head and one tail in any order.

```
# Toss two coins 100 times
coin1 <- sample(c("H", "T"), 100, replace = TRUE)
coin2 <- sample(c("H", "T"), 100, replace = TRUE)

# Combine the outcomes
outcomes <- paste(coin1, coin2, sep = "")

# Frequency of each outcome
table(outcomes)
```

```
## outcomes
## HH HT TH TT
## 19 29 29 23
```

Relative Frequencies

```
# Find the relative frequency
table(outcomes) / 100
```

```
## outcomes
##   HH   HT   TH   TT
## 0.19 0.29 0.29 0.23
```

Probability of One Head and One Tail

One head and one tail can occur as **HT** or **TH**.

```
# Estimate the probability
probability <- sum(outcomes == "HT" | outcomes == "TH") / 100

probability
```

```
## [1] 0.58
```

The estimated probability of getting one head and one tail is given by the simulation result above. It should be close to **0.5**.